

Answer Key — Thinking Like a Scientist

Final answer with a one-line reason. Full methods are in the Notes' worked examples.

Objective

Q1 (b) **Does thicker dough take longer to puff up?** — it can be tested by changing thickness and measuring time.

Q2 (a) **observation** — it is directly seen, with no reasoning added.

Q3 (c) **The puri puffed up in 8 seconds.** — it has a number and a unit.

Q4 (a) **Both A and R true, and R explains A** — "because steam formed" is reasoning, so it is an inference — exactly what R states.

Q5 (b) **False** — science can be done anywhere, even a kitchen.

Q6 **quantitative** — number + unit makes it quantitative.

Q7 **kite** — the kite stands for curiosity taking flight.

Short answer

Q8 **Observation** = what you directly notice (e.g. "the puri is golden"). **Inference** = an explanation from observations (e.g. "it is cooked because it is golden").

Q9 **Qualitative:** "the phulka browned/swelled." **Quantitative:** "it swelled in 4 seconds."

Q10 e.g. "Does hotter oil make a puri puff in less time?" — names what to change and what to measure.

Long / application

Q11 **Question:** "Is the tyre losing air through the valve?" **Change:** apply soapy water to different parts / try a new valve. **Observe/measure:** where bubbles form, or pressure after one night. (Any testable question + one variable + one measurement.)

Q12 **Observations:** "it sank then rose," "it puffed in 6 seconds," "the bottom side was thinner" (any two). **Inference:** "probably because it touched the hot pan first."
